

Knowledge Archive

NHID-Clinical · Public Reference Edition — Vision, Governance & Regulatory Alignment



v1.2

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⚠ NHID-Clinical is an early-stage open proposal by Brianna Baynard. It is not an accredited standard or regulatory requirement. This is a public-safe distillation of the project's internal knowledge archive; internal-only sections (partnership discussions, decision logs, build tasking) are excluded.

Executive Vision

NHID-Clinical was conceived from firsthand experience in payer operations. AI voice agents began appearing in B2B healthcare payer-provider calls — calls that exchange Protected Health Information (PHI) and require human escalation paths — without any consistent disclosure, identity, or audit standard.

Impersonation Latency

The duration of time an AI agent operates and exchanges PHI without disclosing its non-human identity. NHID-Clinical median observation: 3 turns before first disclosure attempt. This is the canonical failure mode the framework exists to prevent.

Gap	Description
Identity Gap	No existing standard requires AI agents to identify themselves before PHI exchange
NPI Authorization Gap	No cross-org NPI delegation mechanism for AI agents calling on behalf of providers
Audit Gap	Call-level AI decisions are not captured in healthcare-compatible audit formats
Escalation Gap	AI agents routinely fail to communicate or honor human escalation requests
Deception Gap	Some vendors use synthetic breathing/hesitation patterns implying human presence

Core Framework — The Four Controls + ATR-01

IDG-01

Identity Disclosure Gate

AI agent must disclose it is automated before any PHI request or exchange.

PDX-01

Pre-Data Exchange Gate

No PHI may be exchanged until IDG-01 disclosure is confirmed.

DBC-01

Deceptive Behavior Check

No synthetic voice artifacts or text claims implying human presence.

EIT-01

Escalation Implementation Test

Human escalation path must be available and honored immediately on request.

ATR-01 · Audit Trail Requirements

Every NHID event must carry event_id, timestamp, session_id, actor_id, state_before/after, and execution_context. Validates as a plain HL7 FHIR R4 AuditEvent (base spec v4.0.1) — no named Implementation Guide conformance is claimed.

Governance — Version Roadmap & CAS

Version	Description	Status
v1.0	Original 4 controls (IDG-01, PDX-01, DBC-01, EIT-01)	Superseded
v1.3	Current: ATR-01 added, CTS expanded to 18 tests, CAS scoring	Current
v2.0	NHID-Auth cryptographic layer (Ed25519, delegation chains)	Reference impl. live
v2.1	Planned: STIR/SHAKEN integration, attestation registry	Future

Identity & Trust Infrastructure

NHID-Auth v2 is the cryptographic authorization layer: provider-signed, NPI-bound agent passports (Ed25519), scoped delegation chains (max 3 hops), per-agent and per-delegation revocation, and call-SID nonce binding. Reference implementation: `src/agent_identity.py`.

Technical Architecture — Live API

Method	Endpoint	Auth	Purpose
POST	/v1/demo/check	none	Raw NHID event → result
POST	/v1/adapters/vapi/check	none	Native VAPI payload → result
POST	/v1/adapters/twilio/check	none	Native Twilio payload → result
POST	/v1/conformance/check	x-api-key	Production conformance check
GET	/health	none	Liveness probe

Regulatory Alignment

Regulatory Driver	Requirement	NHID-Clinical Control
CMS-0057-F	FHIR API, 72hr turnaround, 5yr log	FHIR AuditEvent + ATR-01
MACPAC May 2026	AI transparency, human review	EIT-01 + ATR-01
State AI Laws	Inspectable, auditable decisions	IDG-01 + DBC-01
NIST CAISI 2026	Cross-org agent identity	NHID-Auth v2
HIPAA Security Rule	PHI safeguards + audit controls	PDX-01 + ATR-01
TCPA	Automated caller disclosure	IDG-01

NIST & CMS Reference Standards

Standard	Reference	Relevance
STIR/SHAKEN	RFC 8224 (IETF)	Layer 1: carrier number authentication
FHIR R4	HL7 FHIR v4.0.1	Layer 4: AuditEvent resource
NIST AI RMF	NIST AI 100-1	AI risk management framework

NIST SP 800-207	Zero Trust Architecture	Cross-org authorization patterns
CMS-0057-F	88 FR 80236	Interoperability, FHIR API, claims turnaround
NPI Registry	NPPES (CMS)	10-digit provider identifier

FAQ & Plain Language Guide

Is NHID-Clinical a certification or accreditation?

No. It is a voluntary open proposal (CC BY 4.0). It provides conformance scores, not formal certifications, and has no legal force.

Does NHID-Clinical claim FHIR Implementation Guide conformance?

No. NHID-Clinical validates audit events as plain HL7 FHIR R4 AuditEvent resources against the base spec (v4.0.1) only — it does not claim conformance to any named Implementation Guide such as IHE BALP.

We use a realistic AI voice. Does that automatically trigger DBC-01?

No. DBC-01 evaluates explicit deceptive artifact flags and impersonation phrases — it does not analyze the audio stream directly. Voice realism alone is not a violation.

What does a payer need to start a shadow evaluation?

Access to a sample of incoming administrative call logs, about 30 minutes of staff orientation time, and optional willingness to share anonymized findings.